

Lesson 4-1

Types and Characteristics of Media

Learning Objectives

- Compare the characteristics of different media, and explain the strengths and constraints of websites.
- Choose suitable media for a given purpose and audience, and justify the choice.

Before building a website, it helps to ask: why a website at all? TV reaches millions at once but only one way. Radio can be received almost anywhere but conveys sound only. A newspaper lasts on paper without the Internet. A website combines text, images, and video, updates at any time, and can even respond through forms — while SNS (social networking services) spreads news the fastest of all. Each medium has its own strengths and constraints, so the smart choice depends on the message and the audience.

? Guiding question: What are the strengths and constraints of different media, and how do you choose the right one for a purpose and audience?

Key Fact:
Each medium has different strengths and constraints; choosing the right medium — often several combined — depends on the purpose and the target audience.

Key Concepts:

- TV / radio / newspaper
- website / SNS
- one-way / two-way
- strengths / constraints
- combining media

KEY TERMS
one-way — from sender to receiver only.
two-way — users can respond and share.

Point!

1. Comparison of the Characteristics of Different Media

(1) Each medium for conveying information has different characteristics. When planning a website, it is important to understand the differences from other media.

Medium	Information conveyed	Characteristics
TV	Video + audio	Reaches many people simultaneously. Appeals to sight and hearing. One-way, from sender to receiver.
Radio	Audio	Appeals only to hearing. Can be received almost anywhere and alongside other tasks. One-way.
Newspaper	Text + images	Remains physically as a printed medium. Can be read without an Internet connection. One-way.
Website	Text + images + video + audio	Diverse information can be combined. Can be updated at any time. Basically one-way, but two-way interaction is also possible through forms, comments, etc.
SNS	Text + images + video + audio	Highly effective at spreading information rapidly. Users can post and share information. Two-way.

2. Strengths and Constraints of Websites

(1) Strengths of websites

① Diverse information such as text, images, and video can be combined and presented.

② Information can be updated at any time.

③ Can be accessed from almost anywhere in the world.

④ Two-way interaction with users is possible through forms, comments, etc.

⑤ Can be used with just a browser; no dedicated app installation is required.



Traditional and digital media, side by side.

(2) Constraints of websites

① Normally requires an Internet connection (unless the page is cached).

② Information does not reach users unless they actively access it.

③ Display and loading speed may differ depending on the user's device and network environment.



A website combines strengths one-way media lack.

PAUSE & THINK

Which medium would you choose to reach the most people at once — and which to let them respond to you?

3. Choosing Media According to Purpose

(1) When conveying information, it is important to choose the most suitable medium according to the purpose and target audience.

Purpose	Examples of suitable media	Reason
Inform many people at once	TV, SNS ads	Wide reach
Provide detailed information	Website	No limit on information volume; can be viewed at any time
Deliver information directly to specific people	Email, direct messages	Can be delivered individually
Spread by word of mouth	SNS	Users can voluntarily share and spread information



Match the medium to the purpose and audience.

(2) In practice, multiple media are often used in combination.
e.g., A TV commercial directs viewers to a website, and the website encourages spreading information through SNS.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A A website can present diverse information such as text, images, and video in combination.

B TV is a medium that allows two-way interaction with users.

C SNS is a medium highly effective at spreading information rapidly.

D A website normally can be viewed even without an Internet connection.

(2) Match each purpose (a–c) with the most suitable medium from the choices below (A–C).

a Make detailed information viewable at any time

b Have users spread information by word of mouth

c Inform many people at once with video

[Choices] **A** TV **B** Website **C** SNS

✓ Solution

(1)

A Correct as a strength of websites. Therefore, ○.

B TV is a one-way medium from sender to receiver. Therefore, ×.

C Correct description of the characteristics of SNS. Therefore, ○.

D A website normally requires an Internet connection. Therefore, ×.

(2)

a: A website allows detailed information to be viewed at any time. **B**

b: SNS spreads easily by word of mouth. **C**

c: TV reaches many people with video. **A**

Try

1 Answer the following questions.

1. Name two strengths of websites.
2. Name two constraints of websites.

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT an appropriate characteristic of websites.

A Information can be updated at any time.

B Information automatically reaches users without them actively accessing it.

C Text, images, and video can be combined and presented.

D Two-way interaction is possible through forms and comments.

(2) For each situation (a–d), indicate which medium is most suitable: **A** TV, **B** Website, **C** SNS, **D** Email.

a Make the detailed specifications and usage of a new product available at any time.

b Have users spread information about a new product by word of mouth.

c Deliver video of a new product to many people at once.

d Send campaign information directly to existing customers.

⚙ Think as an Engineer — Investigate & Decide

A public library in your city wants to tell people about a new free reading program. You advise on which media to use.

☐ **Use one medium only** ☐ **Combine media by purpose**

1 · Collect data. Survey ten classmates or family members: where do they actually notice local announcements — a printed poster, Facebook,

EXAM-STYLE QUESTION

Explain why an organization might use a website, TV, and SNS together rather than only one, to promote a new service. **[6] Refer to:** reach, depth, and spreading by word of mouth.

PAUSE & THINK

For each situation below, ask first: is the goal wide reach, full detail, spreading, or a direct message?

WhatsApp, or local radio? Record the results in a table and find the most common channel.

2 · Analyze the audience. For each group — a family with no home Internet, a student, and the librarian — note one benefit and one drawback of using a website as the library's main channel.

3 · Decide. Using your survey and your table, recommend a media plan (which media, in what roles) and give two reasons. **In pairs,** compare your plans and agree together on the single best medium for reaching families who may not be online.

Give evidence for your answer.

Stuck? Start from the audience your survey shows is hardest to reach — which medium already reaches them where they are?

🌍 In a New Context

A hospital wants to inform patients about new opening hours and let them ask questions. Name one medium for reaching many people quickly and one that allows two-way interaction, and say why each one fits.

❓ Lesson Question — Answered

The lesson asked about the strengths and constraints of different media and how to choose. Each medium has its own character: TV reaches many at once but one-way; radio is sound only but reaches anywhere; a newspaper lasts on paper without the Internet; a website combines text, images, and video, updates any time, and allows two-way interaction, though it needs an Internet connection and only reaches those who seek it; and SNS spreads information fastest and is two-way. Because no single medium is best for everything, the right choice depends on the purpose and the target audience — wide reach, full detail, direct delivery, or word-of-mouth — and in practice several media are combined so each does what it does best.

KEY TERMS

website strengths — diverse, updatable, global, two-way.

website constraints — needs Internet; users must seek it.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

Each medium for conveying information has different characteristics. A website has the strengths of being able to present diverse information such as text, images, and video, and being updatable at any time. On the other hand, it has the constraints that it cannot be viewed without an (a) connection, and that information does not reach users unless they actively (b) it. In actual information dissemination, (c) media are often combined, such as directing viewers from a TV commercial to a website.

1. Fill in the blanks (a)–(c).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the characteristics of media.

A Radio is a medium that conveys information through video and audio.

B Newspapers are media highly effective at spreading information rapidly.

C SNS is a medium that allows users to post and share information.

D TV is a medium that allows two-way interaction with users.

(2) From the following options (A–D), choose the one that most appropriately describes the choice of media.

A Information dissemination should always use only one medium.

B Regardless of purpose or target audience, a website is sufficient.

C It is important to choose the most suitable medium according to the purpose and target audience.

D With SNS, TV and websites are not needed.

REFLECT & CHALLENGE

Reflect: Which medium do you personally trust most for important news, and why?
Challenge: Plan the media mix for announcing a school event on a budget that rules out TV — then name one audience your mix might still miss, and why.

Key takeaway

Remember:
No medium is best for everything — match the medium to the purpose and audience, and combine media so each does what it does best.



Lesson 4-2

Information Design and User Experience for Websites

Learning Objectives

- Explain how personas and wireframes are used to plan a website before designing its appearance.
- Apply the four principles of design (contrast, repetition, alignment, proximity) and explain the idea of user-centered design.

A website that makes sense to its designer can still confuse its visitors. Good design therefore begins with the user, not the creator. Designers first picture a real user (a persona), sketch where information will go before choosing any colors (a wireframe), and follow four simple principles — contrast, repetition, alignment, and proximity — so the page is clear at a glance. This user-first mindset is called user-centered design.

? Guiding question: How do designers plan a website around its users, and what principles make a page clear and easy to use?

📌 Point!

1. Persona

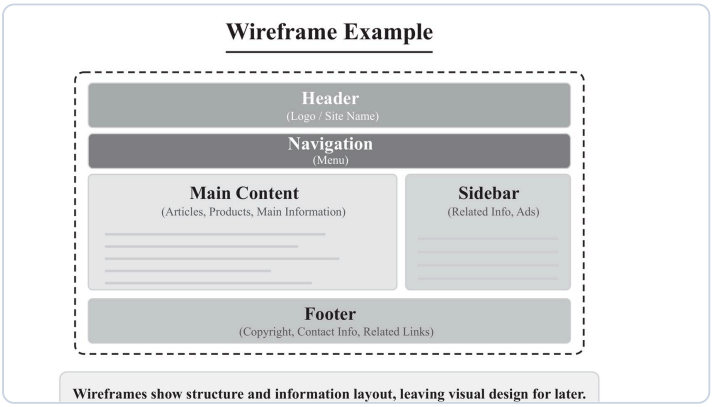
(1) **(Persona)**…a typical user image of those who will use the website, set up as a concrete person. Details such as age, occupation, purpose, and challenges are selected.

(2) By setting up a persona, designers can think specifically about "what this user needs," and the direction of the design becomes less likely to drift.
e.g., Person in their 30s, an office worker, wants to compare products on a smartphone during commute, wants to obtain information in a short time.

2. Wireframe

(1) **(Wireframe)**…a rough blueprint of the layout of a web page. It does not include color or design decoration; it is created to determine the placement and priority of information.

(2) By using a wireframe, "what information to place where" can be considered separately from visual design. If decorations such as colors and fonts are decided first, discussions tend to drift toward personal preferences in appearance, so finalizing the placement and priority of information first makes it easier for stakeholders to reach agreement.



Wireframe Example

Key Fact:
Good web design starts from the user: a persona clarifies who they are, a wireframe fixes what goes where, the CRAP principles make it clear, and user-centered design keeps every choice on the user's side.

Key Concepts:

- persona
- wireframe
- contrast / repetition
- alignment / proximity
- user-centered design

KEY TERMS
persona — a typical user pictured as a concrete person.
wireframe — a layout blueprint before any color or decoration.



A persona captures one target user's goals.

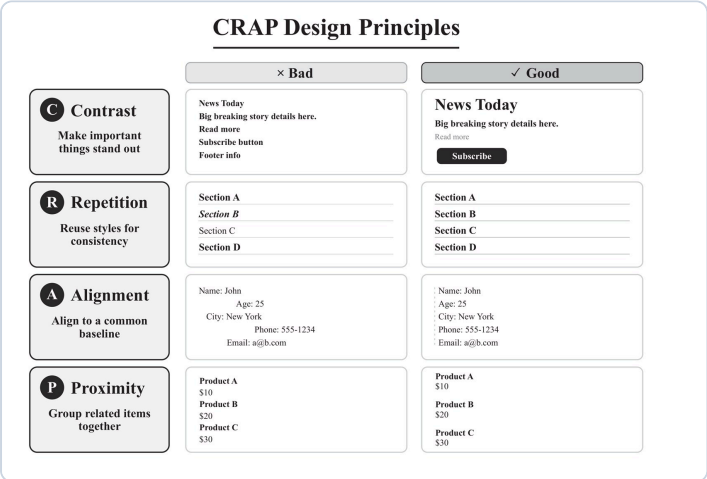


A wireframe grows into a finished design.

3. The Four Principles of Design (CRAP Principles)

(1) Designs that are easy to view and understand follow these four principles.

Principle	Description (purpose)	Specific techniques and examples
(Contrast)	Make the priority of information clear and immediately highlight the most important elements	• Make headings large and bold to differentiate them from body text • Use accent colors for important buttons (e.g., purchase)
(Repetition)	Repeat specific characteristics so that the whole design has consistency, allowing users to navigate without getting lost	• Standardize the color, size, and font of headings on each page • Match the design of bullet point icons
(Alignment)	Align the position of each element to a common baseline, giving order to the entire screen	• Align the left edges of text and images precisely • Make a consistent rule for whether to use center or left alignment
(Proximity)	Place related information close together so that it can be perceived as a single group at a glance	• Group product images, names, and prices together • Leave appropriate spacing (margins) between items



CRAP Design Principles

4. User-Centered Design (UCD)

(1) **(User-Centered Design (UCD))**…a design approach that designs websites and services from the perspective of the user, rather than the convenience of the creator.

(2) In UCD, design begins from “what the user needs” and “in what situation it will be used.” Setting up a persona is one specific way of practicing UCD; by

PAUSE & THINK

Why decide where information goes (wireframe) before choosing colors and fonts?

KEY TERMS

CRAP — contrast, repetition, alignment, proximity.
user-centered design — designing from the user's perspective.

clarifying the user image, design decisions can be kept consistently aligned with the user's perspective.

(3) Improving usability for users and accessibility for a wide range of people are important goals of UCD.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A A persona is a typical user image of those who will use the website, set up as a concrete person.

B A wireframe is a final-version design that includes color and design decoration.

C “Proximity” in the four principles of design means grouping related information close together.

D User-centered design is a design approach that prioritizes the convenience of the creator.

(2) Match each description (a–d) with the most appropriate principle of design from the choices below (A–D).

a Highlight important information and visually indicate priority

b Repeat the same style to give a unified appearance

c Align the positions of elements to give an organized look

d Group related information close together

[Choices] **A** Contrast **B** Repetition **C** Alignment **D** Proximity

✓ Solution

(1)

A Correct definition of a persona. Therefore, ○.

B A wireframe is a blueprint for determining the placement and priority of information, without including color or design decoration. Therefore, ×.

C Correct definition of proximity. Therefore, ○.

D User-centered design is a design approach from the perspective of the user. Therefore, ×.

(2)

a: Highlighting information is contrast. **A**

b: Creating a unified appearance is repetition. **B**

c: Aligning positions is alignment. **C**

d: Grouping related information is proximity. **D**



User-centered design loops around the real user.

PAUSE & THINK

Which principle groups related items together, and which makes the most important item stand out?

Try

1 Answer the following questions.

1. What is the term for a typical user image of those who will use the website, set up as a concrete person?
2. What is the term for a blueprint that shows the placement and priority of information on a web page?
3. What are the four principles of design called when expressed by their English initials?
4. What is the term for the design approach that designs websites and services from the perspective of the user? Also give its three-letter abbreviation.

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT an appropriate piece of information to include in setting up a persona.

A The user's age and occupation

B The user's purpose for using the website

C The performance of the website's server

D The challenges the user faces

(2) For each of design improvements (a–d), indicate which of the four principles of design (A–D) it corresponds to.

a Making headings large and bold to distinguish them from body text

b Grouping product names, prices, and photos close together

c Aligning the left edges of text and images

d Standardizing the font and color of headings across all pages

A Contrast **B** Repetition **C** Alignment **D** Proximity

Think as an Engineer — Investigate & Decide

A community clinic in your city wants a simple website so patients can find opening hours, services, and directions. You lead the design.

☐ **Start by choosing colors and fonts** ☐ **Start from the user, then design**

1 · Collect data. Ask about eight to ten classmates: “If you opened a community clinic's website, what is the one thing you would look for first?” Record the answers in a tally table.

2 · Analyze your users. Group the answers into user types. Include at least one user who is not a typical smartphone visitor — for example an elderly

EXAM-STYLE QUESTION

Explain why a designer creates a persona and a wireframe before choosing colors and fonts. **[6] Refer to:** the user's needs and reaching agreement with stakeholders.

PAUSE & THINK

For each improvement below, ask: does it highlight, unify, order, or group?

patient with low vision, or someone with no home Internet using a library computer. Write what each type needs most.

3 · Decide from your data. Using your tally, decide the priority order of the four home-page blocks and justify each position from your data, not your taste. Then choose one CRAP principle and say exactly how you would apply it for the user type you added. **In pairs,** compare tables and agree on the top block.

Give evidence for your answer.

Stuck? Start with your tally: which need appeared most? Put that block at the top. Then ask of each block — highlight, unify, order, or group? Pick the CRAP principle that fixes the weakest spot for the user you added.

🌍 In a New Context

A school wants to redesign its cluttered results page so parents can find their child's grades quickly. Name the persona you would define, one wireframe decision, and one CRAP principle you would apply, and say what each one improves.

❓ Lesson Question — Answered

The lesson asked how designers plan a website around its users and what makes a page clear. Design begins from the user, not the creator. A persona pictures a typical user as a concrete person — age, occupation, purpose, and challenges — so the team can focus on what that user needs. A wireframe then fixes the placement and priority of information before any color or decoration, which keeps discussion on structure and helps stakeholders agree. To make the page clear, four principles are applied: contrast to highlight priority, repetition for consistency, alignment for order, and proximity to group related items. All of this is guided by user-centered design (UCD), which starts from what the user needs and aims to improve usability and accessibility. Good design, therefore, is planned from the user inward — who they are, what goes where, and how to make it clear.

KEY TERMS

usability — how easily users can use the site.

accessibility — usable by a wide range of people.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

When designing a website, it is important to view it from the user's perspective. First, create a (a), in which a typical user image is set up as a concrete person, and clarify "what this user needs." Next, create a (b), a blueprint for determining the placement and priority of information. In design, to improve viewability and clarity, the (c)—contrast, repetition, alignment, and proximity—should be kept in mind.

1. Fill in the blanks (a)–(c).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT included in the four principles of design (CRAP principles).

A Contrast
B Encryption
C Alignment
D Proximity

(2) From the following options (A–D), choose the one that most appropriately describes the concept of user-centered design.

A Prioritize the design that the developer finds easy to use.
B Place the highest importance on visual design.
C Begin the design from what the user needs.
D Use as many of the latest technologies as possible.

⭐ REFLECT & CHALLENGE

Reflect: On a website you find confusing, which CRAP principle seems most broken, and why? **Challenge:** Your clinic can fund only one accessibility improvement this year. For the low-vision or no-Internet user you identified, choose one CRAP principle to apply — and say what you would give up (a decoration or feature) to make room for it, and why that trade-off serves the user better.

⭐ Key takeaway

Remember:
Persona (who) → wireframe (where) → CRAP (make it clear) → UCD (decide for the user). Design from the user inward.



Lesson 4-3

Methods for Evaluating Websites

Learning Objectives

- Distinguish qualitative from quantitative evaluation, and explain the major indicators used in web analytics.
- Explain how heuristic evaluation works, and why combining both approaches gives a clearer picture of a website.

How do you know whether a website works well for its users? You can watch real users try it and ask what confuses them — that is qualitative evaluation. Or you can count how many people visit, how long they stay, and how many buy — that is quantitative evaluation. Numbers tell you how much is happening; users' voices tell you why. The best teams use both, together with expert checks, to see a site clearly and improve it.

? Guiding question: How can a website be evaluated, and why is it useful to combine numbers with users' voices?

Key Fact:
Websites are evaluated qualitatively (why — usability tests, interviews, heuristic checks) and quantitatively (how much — web analytics: PV, bounce rate, CVR); combining both gives the clearest picture.

Key Concepts:

- qualitative evaluation
- quantitative evaluation
- web analytics
- PV / bounce rate / CVR
- heuristic evaluation

KEY TERMS
qualitative evaluation — the "why" from users' voices and behavior.
quantitative evaluation — the "how much" from numerical data.



Qualitative watches users; quantitative reads numbers.

Point!

1. Qualitative and Quantitative Evaluation

(1) Methods for evaluation generally fall into two main categories: **(qualitative evaluation)** and **(quantitative evaluation)**. These evaluation methods are used in a wide range of applications, and the same concepts apply to website evaluation.

Evaluation method	Description (purpose)	Specific techniques
Qualitative evaluation	Explore in depth the "reasons" behind users' psychology and behavior, and evaluate qualitative aspects that cannot be expressed in numbers.	• Usability testing: observe actual operations to identify points that are hard to use. • Interviews: hear users' direct voices to understand their needs and dissatisfactions.
Quantitative evaluation	Use numerical data such as access counts and time, and evaluate the current state objectively and statistically.	• Web analytics: analyze metrics such as PV count and time spent. • A/B testing: compare two options to verify which leads to better results.

(2) Qualitative evaluation is suited to exploring the cause of "why is it hard to use," while quantitative evaluation is suited to grasping the overall trend of "how much is it being used." By combining both, more accurate evaluation is possible.

PAUSE & THINK

Would counting visits tell you *why* people leave a page? Which method would?

2. Major Indicators in Web Analytics

(1) **(Web analytics)**…a technique for analyzing how a website is accessed using numerical values; it is the central method of quantitative evaluation for websites. The following indicators are commonly used.

Indicator	Meaning	Description
(PV (Page View))	The total number of times a page has been viewed.	A basic indicator that measures how widely the site is seen and how much exposure it gets.
(Bounce rate)	The proportion of visits that ended after viewing only the first page.	An indicator of whether the page content matched the user's expectations and whether further actions were prompted.
(Conversion rate (CVR))	The proportion of visits that achieved a target action (purchase, registration, etc.).	A very important indicator that measures how much the site contributes to business results (profits).

3. Heuristic Evaluation

(1) **(Heuristic evaluation)**…a method in which experts identify problems in a website's UI (user interface) based on rules of thumb (heuristics). It is a form of qualitative evaluation, used along with usability testing and other methods to investigate the causes of usability problems.

(2) Major evaluation perspectives

① Consistency…whether the design and operations are unified across the site.

② Error prevention…whether the design makes it hard for users to make mistakes.

③ Visibility of system status…whether system status is clearly indicated to users (e.g., processing indicators, input error messages).

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Qualitative evaluation is a method that uses numerical data to evaluate quantitatively.

B PV (Page View) is an indicator showing the number of times a page has been viewed.

C The bounce rate is an indicator showing the proportion of visits that achieved a target action.

D Heuristic evaluation is a method in which experts identify UI problems based on rules of thumb.

(2) Match each description (a–c) with the most appropriate indicator from the choices below (A–C).

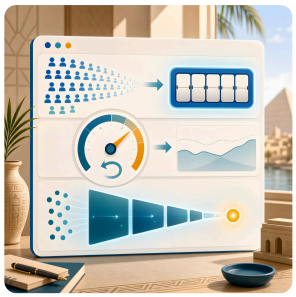
a The number of times a page has been viewed

b The proportion of visits that ended after viewing only the first page

c The proportion of visits that achieved a target action

[Choices] **A** PV (Page View) **B** Bounce rate **C** Conversion rate

✓ Solution



Analytics tracks visits, bounce, and conversions.

KEY TERMS

PV — total page views.
bounce rate — the share of visits that end after only one page.
CVR — the share of visits that reach the target action.



Heuristic review checks a site against usability rules.

PAUSE & THINK

Match the question to the method: “how many?” vs “why?”

(1)

A Qualitative evaluation observes user voices and behavior to evaluate qualitatively. Using numerical data is quantitative evaluation. Therefore, ×.

B Correct definition of PV. Therefore, ○.

C The bounce rate is the proportion of visits that ended after viewing only the first page. The proportion of target action achievement is the conversion rate. Therefore, ×.

D Correct definition of heuristic evaluation. Therefore, ○.

(2)

a: The number of page views is PV. **A**

b: The proportion who left after only the first page is the bounce rate. **B**

c: The proportion who achieved a target is the conversion rate. **C**

Try

1 Answer the following questions.

1. What is the term for the method that evaluates qualitatively by observing user voices and behavior?
2. What is the term for the method that evaluates quantitatively using numerical data?
3. What is the two-letter abbreviation for the indicator showing the number of times a page has been viewed?
4. What is the term for the method in which experts identify UI problems based on rules of thumb?

2 Answer the following questions.

(1) From the following options (A–D), choose all terms that are part of quantitative evaluation methods.

A Usability testing

B Web analytics

C Interviews

D A/B testing

EXAM-STYLE QUESTION

Explain why a website team should use both qualitative and quantitative evaluation rather than only one. **[6] Refer to:** what numbers show and what users' voices show.

PAUSE & THINK

Which of the four above rely on numbers, and which on people's words?

(2) From the following options (A–D), choose the one that most appropriately describes the conversion rate.

A The proportion of times a page has been viewed

B The proportion of visits that ended after viewing only the first page

C The proportion of visits that achieved a target action (product purchase, member registration, etc.)

D The proportion of viewed pages relative to the total number of pages on the site

Think as an Engineer — Investigate & Decide

A small online bookstore in your city wants to know whether its website really works for customers. You plan how to evaluate it.

☐ **Rely on the owner's opinion** ☐ **Use both quantitative and qualitative methods**

1 · Collect data. Ask five classmates who have used an online bookstore to visit any online shop and, for each, record one number you could measure (e.g., how many pages they viewed) and one thing they said was confusing. Put the results in a small table.

2 · Analyze the signals. Suppose the bookstore's PV is high but its conversion rate is very low. What might that mean — and which method (quantitative or qualitative) would tell you *why*? Then consider one customer the data may hide — for example, someone shopping only on a small phone screen, or a first-time visitor — and note one difficulty they might face that a raw PV count would never show.

3 · Decide. Recommend which two evaluation methods the bookstore should use first and give two reasons drawn from the table you built in step 1. **In pairs,** compare your choices and agree together on the single most useful indicator to track.

Give evidence for your answer.

Stuck? Start from the gap in the data: a high PV with a low CVR means people arrive but don't act — so the numbers only point to *where* to look. Pick the qualitative method (a short usability test or an interview) that answers the exact question your numbers raised.

🌍 In a New Context

A school's new website gets many visits, but most people leave after the first page. Name one quantitative indicator that reveals this problem, one qualitative method to find its cause, and one likely fix.

❓ Lesson Question — Answered

The lesson asked how a website can be evaluated and why numbers and users' voices are combined. Evaluation falls into two kinds. Quantitative evaluation uses numerical data — through web analytics indicators such as PV (how many views), bounce rate (how many leave after the first page), and CVR (how many reach a target action) — to show how much a site is used. Qualitative evaluation explores the reasons behind users' behavior through usability testing, interviews, and heuristic evaluation, in which experts check the UI against rules of thumb. Numbers reveal what is happening but not why; users' voices and expert checks reveal the causes. Combining both — the "how much" and the "why" — gives the most accurate evaluation and the clearest path to improvement.

KEY TERMS

web analytics — numbers on how a site is accessed.
heuristic evaluation — experts check the UI by rules of thumb.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

Methods for evaluating a website include (a) evaluation, which observes user voices and behavior, and (b) evaluation, which analyzes using numerical data. Typical indicators used in (b) evaluation are (c), which shows the number of times a page has been viewed; (d), which shows the proportion of visits that ended after viewing only the first page; and (e), which shows the proportion of visits that achieved a target action.

1. Fill in the blanks (a)–(e).

3 Answer the following questions.

(1) For each of evaluation methods (a–c), indicate which it falls under: **A** Qualitative evaluation, **B** Quantitative evaluation.

a Have users actually explore the website and observe its usability

b Tally the access counts and page views of the website

c Interview users to hear their impressions and improvement requests

⭐ REFLECT & CHALLENGE

Reflect: If you could see only one metric for a site you use, would you choose PV, bounce rate, or CVR — and why?
Challenge: A page has a high bounce rate. Design a short qualitative test that would reveal the cause, and say what you would change first.

(2) From the following options (A–D), choose the one that most appropriately describes heuristic evaluation.

A A method in which users explore the site and usability is observed

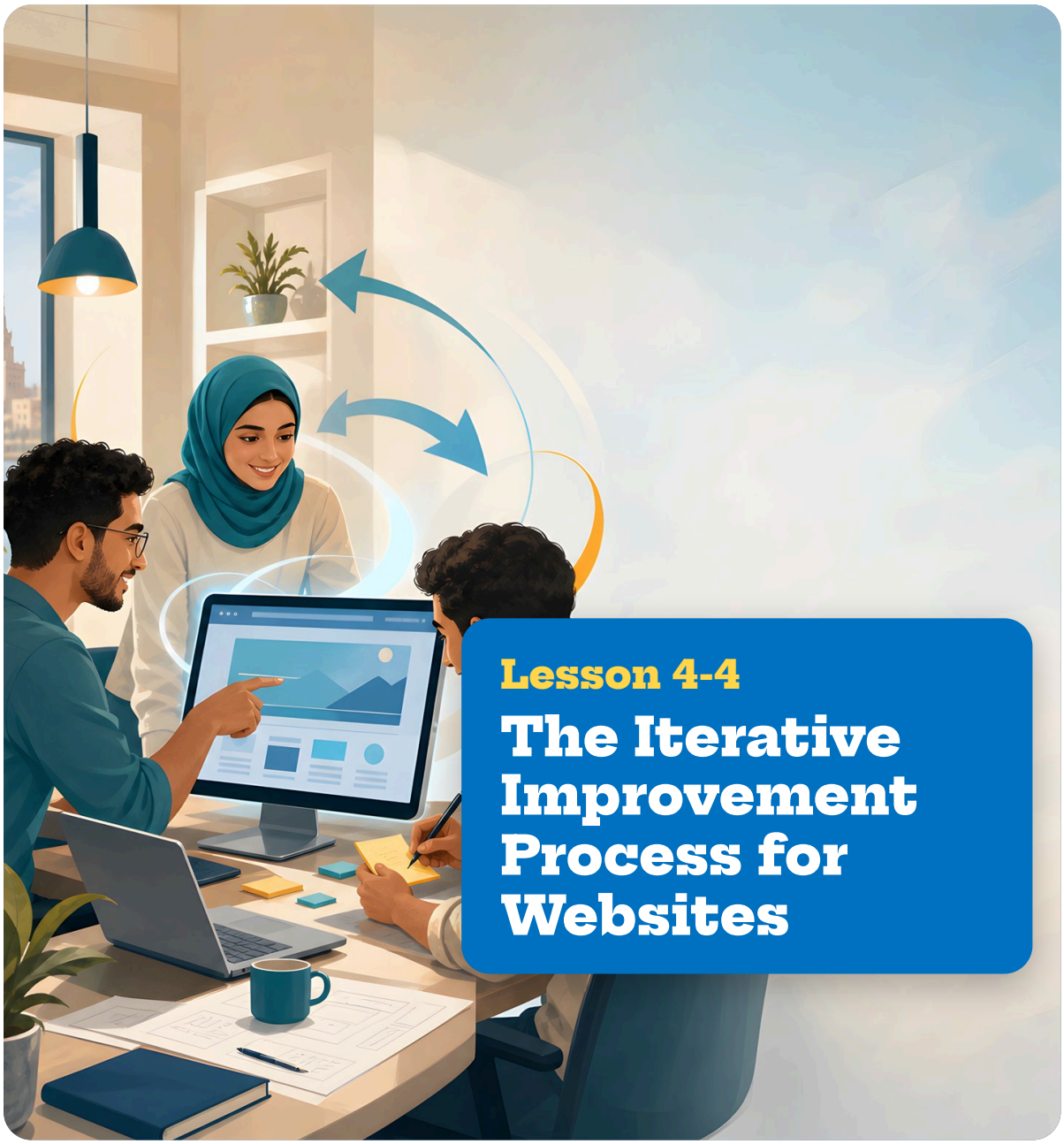
B A method in which numerical data is tallied and site usage is analyzed

C A method in which experts identify UI problems based on rules of thumb

D A method in which two patterns are prepared and compared with data to see which is more effective

Key takeaway

Remember:
Quantitative (PV, bounce rate, CVR) shows how much; qualitative (usability tests, interviews, heuristic checks) shows why. Combine both to evaluate a site clearly.



Lesson 4-4

The Iterative Improvement Process for Websites

Learning Objectives

- Explain the PDCA cycle and the steps of design thinking as ways of improving a website.
- Describe how A/B testing compares two options using real data, and explain why improvement is repeated in cycles.

A website is never truly finished. The first version is only a best guess, and a team learns what truly works only after real people use it. That is why good teams improve in loops: they plan a change, make it, measure the result, and act on what they learn — the PDCA cycle. They also design from the user's side (design thinking) and test real choices with data (A/B testing). Small, repeated improvements are better than waiting for one perfect launch.

? Guiding question: Why is improving a website a repeated loop rather than a one-time job, and how do teams decide what to change?

Point!

1. The PDCA Cycle

(1) **(PDCA cycle)**…a method that continuously advances improvement by repeating the following four steps. It is used widely in fields such as business improvement and quality management, not just websites.

Step	Description	Example in website improvement
Plan	Set goals and create a plan for improvement	Set the goal “reduce bounce rate by 10%” and plan a redesign of the homepage
Do	Execute according to the plan	Change the design of the homepage
Check	Measure results and evaluate the level of goal achievement	Measure the bounce rate after the change and evaluate whether the goal (10% reduction) was achieved
Act	Based on the evaluation, consider the next improvement measures	If the effect is sufficient, continue; if not, consider next steps such as revising headings or image placement

Key Fact:
Websites improve through repeated cycles: PDCA (Plan, Do, Check, Act) drives continuous improvement; design thinking finds problems from the user's side; A/B testing compares two options with real data.

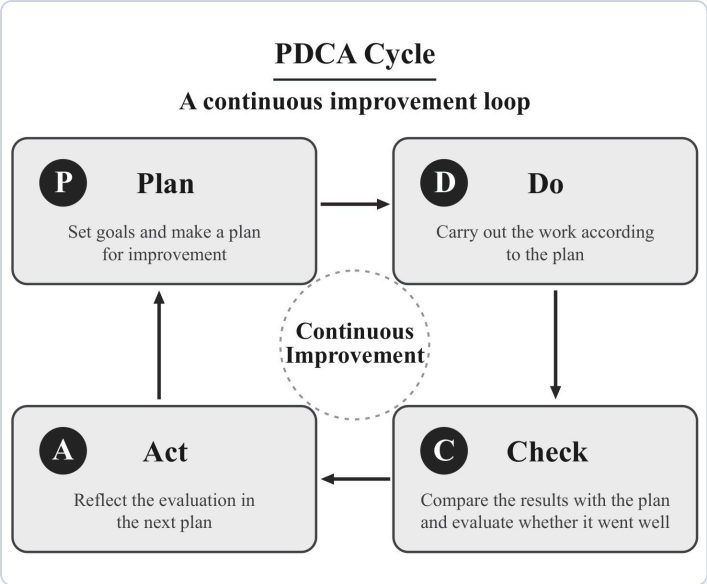
Key Concepts:

- PDCA cycle
- Plan / Do / Check / Act
- design thinking
- A/B testing

KEY TERMS
PDCA cycle — Plan→Do→Check→Act, repeated.
continuous improvement — getting better a little at a time.



Each Plan-Do-Check-Act cycle improves the site.



The PDCA cycle as a continuous improvement loop

(2) The important point of the PDCA cycle is not to aim for completion in one round, but to gradually improve through repetition.

2. Design Thinking

(1) **(Design thinking)**…a way of thinking that leads to discovering problems and generating solutions from the perspective of the user. It proceeds through the following five steps.

Step	Description
① **Empathize**	Put yourself in the user's position and understand their challenges and needs
② **Define**	Based on the information gleaned from empathy, clarify the problem to be solved
③ **Ideate**	Generate many ideas to solve the problem
④ **Prototype**	Quickly create a prototype based on the ideas
⑤ **Test**	Have users try the prototype and obtain feedback from them

(2) Based on test results, one may go back to the empathize or define stage. Improvement proceeds back and forth, not one-way.

PAUSE & THINK

Which PDCA step tells you whether your change actually worked?



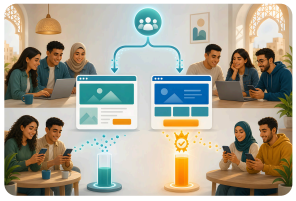
Design thinking: ideate, prototype, test, repeat.

3. A/B Testing

(1) **(A/B testing)**…a method in which two patterns (option A and option B) are prepared, each is shown to actual users, and data is examined to determine which is more effective.

e.g., Compare option A with a red purchase button and option B with a blue purchase button to see which gets clicked more.

(2) Key points of A/B testing



A/B testing compares two versions with real users.

① Change only one element at a time (if multiple elements are changed, it is unclear which change produced the effect).

② Make decisions only after collecting a sufficient amount of data.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A The PDCA cycle is a method that repeats the four steps Plan→Do→Check→Act.

B Design thinking is a way of thinking that solves problems from the developer's perspective.

C In A/B testing, it is important to change only one element at a time.

D The PDCA cycle is a method that is complete after one execution.

(2) For each of a–e, indicate which of the five steps of design thinking (A–E) it corresponds to.

a Understand the user's challenges and needs.

b Clarify the problem to be solved.

c Generate many ideas.

d Quickly create a prototype.

e Have users try the prototype.

A Empathize B Define C Ideate D Prototype E Test

✓ Solution

(1)

A Correct definition of the PDCA cycle. Therefore, ○.

B Design thinking is a way of thinking from the user's perspective. Therefore, ×.

C It is important to limit the changed element to one in A/B testing. Therefore, ○.

D The PDCA cycle is meaningful when repeated. Therefore, ×.

(2)

a: Understanding the user is empathize. **A**

b: Clarifying the problem is define. **B**

c: Generating ideas is ideate. **C**

d: Creating a prototype is prototype. **D**

e: Having users try is test. **E**

KEY TERMS

design thinking — solve from the user's view in five steps.

A/B testing — compare two options with real data.

💡 PAUSE & THINK

In A/B testing, why change only one element at a time?

Try

1 Answer the following questions.

1. What is the term for the improvement method that repeats the four steps Plan→Do→Check→Act?
2. What is the term for the way of thinking for discovering problems and generating solutions from the perspective of the user?
3. What is the term for the method in which two patterns are prepared and compared with data to see which is more effective?
4. Name the five steps of design thinking in order.

2 Answer the following questions.

(1) From the following options (A–D), choose the one that corresponds to "Check" in the PDCA cycle.

A Set the goal of reducing the bounce rate by 10%.

B Change the design of the homepage.

C Measure the bounce rate after the change.

D Consider another improvement measure.

(2) From the following options (A–D), choose the one that is NOT an appropriate way of conducting A/B testing.

A Prepare two patterns and compare them.

B Change multiple elements at the same time.

C Make decisions only after collecting a sufficient amount of data.

D Show each pattern to actual users.

Think as an Engineer — Investigate & Decide

Your school club runs a web page for its yearly event, but few students sign up through it. You lead one round of improvement using the PDCA cycle.

☐ **Redesign everything at once** ☐ **Improve in one measured PDCA loop**

1 · Collect data (Check first). Decide one number you could measure now — for example, how many of ten classmates find the "sign up" link within five seconds. Record the result as your starting point.

2 · Plan one change. From the users you watched, name the single biggest obstacle (for a specific group, such as a student opening the page on a small phone screen) and plan one change to fix it. State the goal as a number (e.g., "8 of 10 find it in five seconds").

EXAM-STYLE QUESTION

Explain why a website team improves in repeated PDCA cycles instead of trying to build a perfect site in one go. **[6]**
Refer to: measuring results and acting on them.

PAUSE & THINK

Match each option above to a PDCA step: which is Plan, Do, Check, Act?

3 · Do, Check, Act — and decide. Describe how you would make the change, measure again, and decide the next step. Choose one element to settle with an A/B test, and say why you change only that one element. **In pairs,** compare plans and agree on the single change to try first.

Give evidence for your answer.

Stuck? Start from your measured number (Check), pick the one obstacle it exposes, plan a single fix (Plan), then loop — do not change everything at once.

🌍 In a New Context

An online shop finds that its “buy” button is rarely clicked. Describe how you would use one A/B test (changing only one element) and one PDCA loop to improve it, and say which number would tell you it worked.

❓ Lesson Question — Answered

The lesson asked why improving a website is a repeated loop and how teams decide what to change. A first version is only a guess, so teams improve continuously rather than once. The PDCA cycle repeats four steps — Plan (set a goal and plan a change), Do (make it), Check (measure the result), and Act (decide the next step) — and its point is not completion in one round but gradual improvement through repetition. To decide what to change, design thinking works from the user's side through five steps (Empathize, Define, Ideate, Prototype, Test), moving back and forth rather than one-way, while A/B testing compares two options with real data, changing only one element at a time and deciding only after enough data. Together they make improvement small, measured, and repeatable.

KEY TERMS

prototype — a quick, testable draft.

iterate — repeat the loop to improve.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

As a method for improving websites, there is the (a) cycle. It repeats four steps: (b), in which goals are set; (c), in which the plan is executed; (d), in which results are measured; and (e), in which improvement measures are considered. Also, (f), in which two patterns are prepared and compared with data to see which is more effective, is a useful method.

1. Fill in the blanks (a)–(f).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes design thinking.

A A method that uses numerical data to analyze website usage.

B A way of thinking that leads to discovering problems and generating solutions from the perspective of the user.

C A method in which experts identify UI problems based on rules of thumb.

D A method of creating a blueprint that roughly shows the layout of a web page.

(2) From the following options (A–D), choose the one that most appropriately describes the concept of the website improvement process.

A If a perfect website is built initially, no improvement is needed.

B It is enough to improve only the visual design.

C It is important to gradually improve by repeating evaluation and improvement.

D All improvements should be done with A/B testing alone.

⭐ REFLECT & CHALLENGE

Reflect: Think of a website you use often. Which one small change do you think would help most, and how would you measure whether it worked?
Challenge: You may run only one A/B test this month on the event page. Choose the single element to test, predict which option will win and why, and say what number would prove you right or wrong.

⭐ Key takeaway

Remember:
Improve in loops: PDCA (Plan, Do, Check, Act) for continuous improvement, design thinking (Empathize, Define, Ideate, Prototype, Test) from the user's side, and A/B testing to settle one change with data.